30-Day Roadmap & Costing

Four-week pilot plan, post-pilot rollout, one-time & monthly running costs, staffing, strict-NDA sizing, and risk controls

30-Day Pilot Roadmap (Weeks 1–4)

Phase	Key Activities	Exit / Gate Criteria
Week 1 · Baseline & Permissions	Shadow QC team on live sets; log QC hours/set & escaped RFIs; collect HR cost rates; secure client consent for 3–5 past projects; lock NDA & cloud-AI rules.	Baseline signed off; written data consent; measured inputs replace ROI placeholders.
Week 2 · Adapt & Back-Test	Configure title-block, tag regex & schedule column maps to Uppteam CAD templates; run on 3–5 completed projects; score vs historical QC logs (aecdi score-qclog).	Recall ≥ 80% on covered issue types; 0 crashes on real PDFs; 50-sheet extraction check.
Week 3 · Live Shadow Pilot	Run in parallel with normal QC on 2 live projects (folder watcher + Streamlit Review UI); engineers Accept / Reject (with reason) every finding; nothing sent to clients.	Zero delivery disruption; live reviewer acceptance rate (precision) measured.
Week 4 · Calibrate & Go/No-Go	Convert rejections to regression tests (aecdi feedback / regress); time sets with vs without tool; hold QC workshop; recompute ROI from measured Week 1–3 data.	GO if: Precision ≥ 85%, Recall ≥ 80%, QC time cut ≥ 30%, and capacity plan agreed.

Pilot & Steady-State Cost Breakdown

Cost Component	Amount (USD)	Basis & Assumptions (FX: ₹88/$)
30-Day Pilot (One-Time)	$2,769	100% AI Lead ($2,273) + 25% QC Specialist ($256) + 10% IT ($91) + Cloud/LLM ($150)
Cloud Infrastructure (Monthly)	$100 / mo	4-vCPU / 16 GB VM ($60) + PostgreSQL, storage & backups ($40); or existing office server
LLM API Usage (Monthly)	$34.50 / mo	30 sets × $1.15/set (25 spec sections/set, 4k in / 1.5k out tokens, Claude Opus 5.5 list)
People Operations (Monthly)	$556.82 / mo	20% AI Lead ($454.55) + 10% Senior QC Specialist ($102.27) for tuning & QA
Total Steady-State Running Cost	$691 / mo	$134.50 Cloud & LLM + $556.82 staff allocation (all at ₹88/$ exchange rate)

Staffing Allocation (Pilot vs. Steady State)

Role	India CTC	Pilot	Steady	Primary Responsibility
AI Process Optimization Lead	₹2,00,000 / mo	100%	20%	Rule tuning, CAD onboarding, evaluation, integrations
Senior QC Specialist	₹90,000 / mo ($5.81/h)	25%	10%	Ground-truth QC logs, shadow review, feedback triage
IT Administrator	₹80,000 / mo	10% (Wk 1)	On demand	VM provisioning, folder permissions, NDA controls

Post-Pilot Rollout (If Week-4 Gate Passes)

Window	Focus	Target Milestone
Days 31–60 · Team Rollout	Extend to 2–3 production pods; enable ACC posting of accepted findings on 1 client project; add next CAD standard.	Accept rate holds ≥ 85% across pods
Days 61–90 · Second Eye Pilot	Validate pricing, contract scope ('coordination review, no stamp') & liability cover; offer AI-assisted QC to 1–2 existing clients.	Signed pilot client or formal no-go
Months 4–6 · Production	PostgreSQL backend, SSO, OCR branch for raster scans, IFC / pyRevit direct export integration.	Unattended runs on all new drawing issues

Strict-NDA / Offline Deployment Options

Profile	Hardware	Cost	Capability
Deterministic Rules Only	Standard CPU (no GPU, no internet)	$0 extra	Runs 32 deterministic checks 100% offline
Local LLM · 24 GB GPU	Workstation (e.g. RTX 4090)	One-time capex	Fits ~30B 4-bit open-weight models offline
Local LLM · 48 GB GPU	Workstation (2×24 GB or A6000)	Higher capex	Fits ~70B 4-bit models (~40 GB weights)
Private Cloud GPU	AWS g5.xlarge (24 GB)	~$1/hr on-demand	Started only during batch runs (~$15–$50/mo)

Key Operational & Commercial Risks

Risk	Impact / Prob.	Mitigation Built Into Plan
Real CAD standards differ from synth	High / High	Week-2 adaptation sprint; per-client TagStyle config; raster guard
Reviewer over-reliance on tool	High / Med	Publish covered vs uncovered matrix; track missed-issue rate
Saved QC hours not redeployed	High / Med	Week-4 Go/No-Go requires management capacity plan sign-off
Client NDA / cross-border data	High / Med	Cloud LLM off by default; send only spec text with zero retention
PyMuPDF AGPL if offered externally	Med / High	Acquire commercial PyMuPDF licence before external SaaS/client use

Low upfront commitment

$2,769 one-time pilot cost is 95% existing staff time; only $150 is cash outlay for cloud and API testing.

Offline by default

All 32 deterministic rules run locally on CPU with zero cloud calls; LLM is optional and restricted to spec text.

Governance & liability

Tool outputs are "potential discrepancies" only. The Engineer of Record decides; nothing auto-publishes to ACC unaccepted.

Hard Week-4 Go/No-Go

No production rollout unless real-project Recall ≥ 80%, Precision ≥ 85%, and QC time reduction ≥ 30% are proven.